exercise 1

$S = -\sum_{G \in M_n} P(G) \ln P(G)$ $\qquad M_n = \mathbb{M}^{N \times N}(\{0,1\})$

we want to find probability distribution maximizing entropy subject to constraints $\sum_G P(G) x_i(G) = \langle x_i \rangle$

$\{x_i\}$ - collection of graph observables that we have measured in empirical observation

$\sum_G P(G) = 1$ normalization condition

$x_i(G)$ - value of x_i in graph G

estimate $\langle x_i \rangle$ of expectation value of each observable

$$\mathcal{L} = -\sum_G P(G) \ln P(G) + \alpha\left(1 - \sum_G P(G)\right) + \sum_i \theta_i \left(\langle x_i \rangle - \sum_G P(G) x_i(G)\right)$$

$$\frac{\partial \mathcal{L}}{\partial P(G)} = -\ln P(G) - 1 - \alpha - \sum_i \theta_i \cdot x_i(G)$$

$$\ln P(G) + 1 + \alpha + \sum_i \theta_i x_i(G) = 0$$

$$\ln P(G) = -(1 + \alpha + H(G))$$

$$P(G) = e^{-1-\alpha} \cdot e^{-H(G)}$$

$H(G) = \sum_i \theta_i x_i(G)$

$$\sum_G e^{-1-\alpha} e^{-H(G)} = 1$$

$$e^{-1-\alpha} = \frac{1}{\sum_G e^{-H(G)}}$$

$$e^{\alpha+1} = \sum_G e^{-H(G)}$$

$Z = e^{\alpha+1} = \sum_G e^{-H(G)}$ ← partition function

$$P(G) = \frac{e^{-H(G)}}{\sum_G e^{-H(G)}}$$

$H(G) = \sum_i \theta_i x_i(G)$ - graph Hamiltonian

(The statistical mechanics of networks Juyong Park and M.E.J. Newman, 2004)

p 5.7.

First the partition function

$H(G) = \theta m(G)$ $\qquad \langle m \rangle = \sum_G P(G) m(G)$

$$\sigma_{ij} = \begin{cases} 1 & \text{if } i \text{ is connected to } j \\ 0 & \text{otherwise} \end{cases}$$

← adjacency matrix

The number of edges $m = \sum_{i<j} \sigma_{ij}$

$$Z = \sum_G e^{-H} = \sum_{\{\sigma_{ij}\}} \exp\left(-\theta \sum_{i<j} \sigma_{ij}\right) = \sum_{\{\sigma_{ij}\}} \prod_{i<j} e^{-\theta \sigma_{ij}} =$$

$$= \sum_{\sigma_{12}=0}^{1} \sum_{\sigma_{13}=0}^{1} \dots \sum_{\sigma_{(n-1)n}=0}^{1} \prod_{i<j} e^{-\theta \sigma_{ij}} =$$

$$= \prod_{i<j} \sum_{\sigma_{ij}=0}^{1} e^{-\theta \sigma_{ij}} = \prod_{i<j} (1 + e^{-\theta}) = [1 + e^{-\theta}]^{\binom{N}{2}}$$

← number of possible pairs i,j $i<j$

$$-\ln Z = -\binom{n}{2} \ln(1 + e^{-\theta})$$

$$\langle m \rangle = \sum_G P(G) m(G) = \sum_G \frac{e^{-H(G)}}{Z} \cdot m(G) =$$

$$= \frac{1}{Z} \sum_G e^{-\theta m(G)} m(G) = -\frac{1}{Z} \frac{\partial Z}{\partial \theta} = -\frac{\partial \ln Z}{\partial \theta} = \binom{n}{2} \frac{1}{e^{\theta} + 1}$$

$p = \frac{1}{e^{\theta}+1}$ $\qquad \langle m \rangle = \binom{n}{2} p$

$$P(G) = \frac{e^{-H}}{Z} = \frac{e^{-\theta m}}{[1 + e^{-\theta}]^{\binom{n}{2}}} = p^m (1-p)^{\binom{n}{2} - m}$$

exercise 5

any vertex v can be connected to $N-1$ other vertices

$N-1$ independent trials

edge exists with probability p
edge doesn't exist with probability $1-p$

for a degree distribution we want sum of when edge exist,
which is a binomial distribution

$$\mathbb{P}(\deg(v)=k) = \binom{N-1}{k} p^k (1-p)^{(N-1-k)}$$

The expected value in binomial distribution is
$(N-1)\cdot p$

exercise 6

I'll show what happens when $N \to \infty$ and mean stays finite

let $\deg(v)_n \sim \text{Binomial}(n-1, p_n)$ with
if $n \to \infty$ then $p_n \to 0$ $(n-1)p_n = \lambda > 0$ (mean stays finite)
if it needs to stay λ

we want to show $\lim_{n \to \infty} \mathbb{P}(\deg(v)_n = k) = \frac{\lambda^k e^{-\lambda}}{k!}$

$$\binom{n-1}{k} p_n^k (1-p_n)^{n-1-k} =$$

$$\binom{n-1}{k} = \frac{(n-1)\dots(n-k)}{k!} = \frac{(n-1)^k}{k!} \cdot \frac{(n-1)\dots(n-k)}{(n-1)^k} =$$

$$\varepsilon(n) \xrightarrow[n\to\infty]{} 0$$

$$= \frac{(n-1)^k}{k!} \cdot 1 \cdot \left(1-\frac{1}{n-1}\right)\dots\left(1-\frac{k-1}{n-1}\right) = \frac{(n-1)^k}{k!}(1+\varepsilon(n))$$

$$\binom{n-1}{k} p_n^k = \frac{(n-1)^k}{k!} \cdot (1+\varepsilon(n)) \cdot \frac{[(n-1)p_n]^k}{(n-1)^k} = \frac{[(n-1)p_n]^k}{k!}$$

$$\cdot (1+\varepsilon(n)) \xrightarrow[n\to\infty]{} \frac{\lambda^k}{k!}$$

$$(1-p_n)^{n-1-k} = \exp[(n-1-k)\ln(1-p_n)]$$

$$\ln(1-p_n) = -p_n - \frac{p_n^2}{2} + O(p_n^3)$$ taylor's expansion

$$(n-1-k)\ln(1-p_n) = -(n-1-k)p_n - \frac{(n-1-k)p_n^2}{2} + (n-1-k)O(p_n^3)$$

$$(n-1-k)p_n = (n-1)p_n - kp_n \to \lambda - 0 = \lambda$$

$$(n-1-k)p_n^2 = (n-1)p_n^2 - kp_n^2 \to 0$$

$$(n-1-k)p_n^3 = (n-1)p_n^3 - kp_n^3 \to 0 \qquad |O(p_n^3)| \le C \cdot p_n^3$$

so

$$(n-1-k)\ln(1-p_n) \to -\lambda \Rightarrow (1-p_n)^{n-1-k} \to e^{-\lambda}$$

$$\binom{n-1}{k} p_n^k \cdot (1-p_n)^{n-1-k} \xrightarrow[n\to\infty]{} \frac{\lambda^k}{k!} e^{-\lambda} \Leftarrow \text{Poisson}$$

p 5.1

$K \sim Poiss(\langle k \rangle)$

$$P(K=k) = \frac{e^{-\langle k \rangle}\langle k \rangle^k}{k!} \qquad k = 0, 1, 2, \dots$$

$$E[K] = \sum_{k=0}^{\infty} k P(K=k) = \sum_{k=0}^{\infty} k \cdot e^{-\langle k \rangle} \cdot \frac{\langle k \rangle^k}{k!} =$$

$$= e^{-\langle k \rangle} \sum_{k=0}^{\infty} k \cdot \frac{\langle k \rangle^k}{k!} = e^{-\langle k \rangle} \sum_{k=1}^{\infty} \frac{\langle k \rangle^k}{(k-1)!} =$$

$$= e^{-\langle k \rangle} \cdot \langle k \rangle \cdot \sum_{k=1}^{\infty} \frac{\langle k \rangle^{k-1}}{(k-1)!} \underset{m=k-1}{=} e^{-\langle k \rangle} \cdot \langle k \rangle \cdot \sum_{m=0}^{\infty} \frac{\langle k \rangle^m}{m!} =$$

$$= e^{-\langle k \rangle} \cdot \langle k \rangle \cdot e^{\langle k \rangle} = \langle k \rangle$$

$$E[K^2] = \sum_{k=0}^{\infty} k^2 e^{-\langle k \rangle} \frac{\langle k \rangle^k}{k!} = e^{-\langle k \rangle} \sum_{k=0}^{\infty} \frac{k^2 \cdot \langle k \rangle^k}{k!} =$$

$$= e^{-\langle k \rangle} \cdot \left(\sum_{k=0}^{\infty} \frac{k(k-1)\langle k \rangle^k}{k!} + \sum_{k=0}^{\infty} \frac{k \langle k \rangle^k}{k!} \right) =$$

$$= \langle k \rangle + e^{-\langle k \rangle} \cdot \sum_{k=2}^{\infty} \frac{\langle k \rangle^k}{(k-2)!} = \langle k \rangle + e^{-\langle k \rangle} \cdot \sum_{k=2}^{\infty} \frac{\langle k \rangle^2 \langle k \rangle^{k-2}}{(k-2)!} =$$

$$= \langle k \rangle + \langle k \rangle^2 \cdot e^{-\langle k \rangle} + \sum_{m=0}^{\infty} \frac{\langle k \rangle^m}{m!} = \langle k \rangle + \langle k^2 \rangle e^{-\langle k \rangle} e^{\langle k \rangle} =$$

$m = k-2$

$$= \langle k \rangle + \langle k^2 \rangle$$

$$var[K] = E[K^2] - [E[K]]^2 = (\langle k \rangle^2 + \langle k \rangle) - \langle k \rangle^2 = \langle k \rangle$$

p 5.5

A graph chosen uniformly at random from all graphs with exactly N nodes and E edges

$M = \binom{N}{2}$ ← possible edges in a graph with N nodes

$P(G) = \frac{1}{\binom{M}{E}}$ ← probability that particular graph with E edges is chosen

from handshaking lemma $\sum_{i=1}^{N} k_i = 2E$

$E[\sum_{i=1}^{N} k_i] = E[2E]$

$\sum_{i=1}^{N} E[k_i] = 2E$

$N \cdot \langle k \rangle = 2E \Rightarrow \langle k \rangle = \frac{2E}{N}$

expected degree

$\langle k \rangle = \frac{2E}{N}$

each node can connect to N−1 other nodes

N−1

edges between other vertices $M-(N-1) = \binom{N}{2} - (N-1)$

so first we choose k edges from those that it can connect to $\binom{N-1}{k}$ k-degree of a node

then E−k from others $\binom{M-(N-1)}{E-k}$

we have $\binom{M}{E}$ possible combinations so

$P(k) = \left[\binom{N-1}{k} \cdot \binom{M-(N-1)}{E-k}\right] / \binom{M}{E}$

hypergeometric distribution $(M, N-1, E)$ of node degrees

so degree variance is

$Var[X] = E \cdot \frac{N-1}{M} \cdot \frac{M-(N-1)}{M} \cdot \frac{M-E}{M-1}$

I'll check mean if it's the same

$E[X] = E \cdot \frac{N-1}{M} = E \cdot \frac{N-1}{\binom{N}{2}} = E \cdot \frac{N-1}{\frac{N(N-1)}{2}} = E \cdot \frac{2}{N} = \frac{2E}{N}$

25/11/2025 homework

networksciencebook.com / chapter / 5# degree-dynamics

p 6.1

$$\Pi(i) = \frac{1}{t+m_0} \approx \frac{1}{t}$$ ← probability that a fixed existing node receives any particular edge at time t

The expected increase of k_i per t is (each step introduces m edges)

$$\frac{dk_i}{dt} = m \cdot \Pi(i) \approx \frac{m}{t}$$

$k_i(t_i) = m$ ← node i appears at t_i and has initially m degrees

$$\frac{dk_i}{dt} = \frac{m}{t}$$

$$dk_i = \frac{m}{t} dt$$

$$\int_{k_i(t_i)}^{k_i(t)} dk_i = \int_{t_i}^{t} \frac{m}{t} dt$$

from (1) & (2)

$$k_i(t) - m = m \cdot \ln \frac{t}{t_i}$$

$$k_i(t) = m \cdot \ln \frac{t}{t_i} + m$$

$$\int_{k_i(t_i)}^{k_i(t)} dk_i = \int_{m}^{k_i(t)} dk_i = k_i(t) - m \quad (1)$$

$$m \cdot \int_{t_i}^{t} \frac{1}{t} dt = m \cdot (\ln t - \ln t_i) = m \cdot \ln \frac{t}{t_i} \quad (2)$$

now the probability

$$k = m \cdot \ln \frac{t}{t_i} + m \Rightarrow k - m = m \ln \frac{t}{t_i}$$

$$\ln \frac{t}{t_i} = \frac{k-m}{m} \Rightarrow \frac{t}{t_i} = e^{(k-m)/m} \Rightarrow t_i = t \cdot e^{-(k-m)/m}$$

$p_{t_i}(t_i) \approx \frac{1}{t}$ all nodes are equally likely to have any birth time

Fact

If we have a random variable X with PDF $p_X(x)$ and $Y = f(X)$ is a monotone function, then the PDF of Y is

$$p_Y(y) = p_X(x) \cdot \left| \frac{dx}{dy} \right|$$

$X = t_i \quad Y = k$ so

$$P(k) = p_{t_i}(t_i) \cdot \left| \frac{dt_i}{dk} \right|$$

$$t_i(k) = t e^{-(k-m)/m} \Rightarrow \frac{dt_i}{dk} = t \cdot e^{-(k-m)/m} \cdot \left(-\frac{1}{m}\right) = -\frac{t_i}{m}$$

$$P(k) = \frac{1}{t} \cdot \left| -\frac{t_i}{m} \right| = \frac{1}{t} \cdot \frac{t_i}{m} = \frac{1}{t} \cdot t \cdot e^{-(k-m)/m} \cdot \frac{1}{m} =$$

$$= e^{-(k-m)/m} \cdot \frac{1}{m} = e^{-k/m} \cdot e^{m/m} \cdot \frac{1}{m} = \frac{e}{m} \cdot e^{-k/m}$$

p 6.2

the expected increase of k_i per t

$$\frac{dk_i}{dt} = \frac{N-1}{N} \frac{k_i}{2t} + \frac{1}{N}$$

$\frac{1}{N}$ - because even node with degree 0 has a chance to be connected

$\frac{k_i}{2t}$ - like in the normal BA

$\frac{N-1}{N}$ - but we need to correct it because we don't consider connecting node to itself

$$\frac{dk_i}{dt} - \frac{N-1}{N} \frac{k_i}{2t} = \frac{1}{N}$$

$$\frac{dy}{dt} + P(t) \cdot y = Q(t)$$

$y = k_i(t)$

$P(t) = -\frac{N-1}{2Nt}$

$Q(t) = \frac{1}{N}$

integrating factor

Simon Fraser University
First Order Linear Differential Equations

$t \geq 0$

$\mu(t) = \exp\left(\int P(t)dt\right) =$

$= \exp\left(\int -\frac{N-1}{2Nt}dt\right) =$

$= \exp\left(-\frac{N-1}{2N}\int \frac{1}{t}dt\right) = \exp\left(-\frac{N-1}{2N}\ln t\right) = t^{-(N-1)/(2N)}$

Multiply ODE by $\mu(t)$

$$t^{-(N-1)/(2N)} \cdot \left(\frac{dk_i(t)}{dt} - \frac{N-1}{2Nt}\cdot k_i(t)\right) = \frac{1}{N}\cdot t^{-(N-1)/(2N)}$$

$$t^{-(N-1)/(2N)} \cdot \frac{dk_i(t)}{dt} - \frac{N-1}{2Nt}\cdot t^{-(N-1)/(2N)} \cdot k_i(t) = \frac{1}{N}\cdot t^{-(N-1)/(2N)}$$

$$\frac{d}{dt}\left(t^{-(N-1)/(2N)}k_i(t)\right) = \frac{1}{N}\cdot t^{-(N-1)/(2N)} \quad \Big| \int$$

$$t^{-(N-1)/(2N)} \cdot k_i(t) = \frac{1}{N}\int t^{-(N-1)/(2N)}dt + C \quad (*)$$

$$\int t^{-(N-1)/(2N)}dt = t^{1-(N-1)/(2N)} \cdot \frac{1}{1-(N-1)/(2N)} + \text{const} =$$

$$= t^{(N+1)/(2N)} \cdot \frac{2N}{N+1} + \text{const}$$

$$(*) \quad t^{-(N-1)/(2N)} \cdot k_i(t) = \frac{1}{N} t^{(N+1)/(2N)} \cdot \frac{2N}{N+1} + C$$

$$k_i(t) = \frac{1}{N}\cdot t^{(N-1)/(2N)} \cdot t^{(N+1)/(2N)} \cdot \frac{2N}{N+1} + Ct^{(N-1)/(2N)}$$

$$k_i(t) = \frac{1}{N}\cdot\frac{2N}{N+1}\cdot t + Ct^{(N-1)/(2N)}$$

$$k_i(t) \approx \frac{2}{N}\cdot t$$

$Ct^{(N-1)/(2N)} \sim Ct^{1/2}$ approx for large N

but $Ct^{1/2}$ is small compared to the linear growth t for large t

p 6.5

network has N nodes (fixed)
edges are added over time using preferential attachment

$k_i(t)$ degree of node i at time t

$k_i(t) \approx \frac{2}{N}t$ each node's degree grows linearly with time

$t(k) \approx \frac{N}{2}k$ effective time corresponding to a node having degree k

let's define t_i as the effective birth time of a node - the time when it starts gaining edges.

the distribution of these effective birth times is

$p(t_i) = \text{uniform on } [0,T] \Rightarrow p(t_i) = \frac{1}{T}$ because all nodes are equivalent

so considering those effective birth times

$k \approx \frac{2}{N}(t - t_i) \Rightarrow t_i = t - \frac{N}{2}k \qquad \frac{dt_i}{dk} = -\frac{N}{2}$

$P(k) = p(t_i)\cdot\left|\frac{dt_i}{dk}\right| = \frac{1}{T}\cdot\frac{N}{2} = \frac{N}{2T}$ so the distribution is flat

Project 7

p 7.4 1P

Let G be a simple graph with N vertices. Let

$$P(k) = \frac{\#\{\text{vertices of degree } k\}}{N}$$ be the degree distribution,

and let $\langle k \rangle = \sum_k k P(k)$ be the mean degree

We assume that the graph is uncorrelated: picking an edge uniformly at random and then picking one of its two ends uniformly is equivalent to picking a half-edge (stub) uniformly at random.

We want to show that:

If $Q(k)$ is the probability that the end-vertex of a uniformly chosen edge has degree k, then

$$Q(k) \propto k P(k) \quad \text{and} \quad Q(k) = \frac{k}{\langle k \rangle} P(k)$$

Proof

The number of vertices of degree k is $NP(k)$. Each such vertex contributes k stubs, hence the total number of stubs attached to vertices of degree k is

$$S_k = k \cdot N \cdot P(k)$$

The total number of stubs in the graph is

$$S = \sum_k S_k = \sum_k k N P(k) = N \sum_k k P(k) = N \cdot \langle k \rangle$$

Choosing an edge uniformly at random and then selecting one of its two ends uniformly is equivalent to choosing a stub uniformly at random. Therefore

$$Q(k) = \frac{S_k}{S} = \frac{k \cdot N P(k)}{N \langle k \rangle} = \frac{k \cdot P(k)}{\langle k \rangle} \quad \text{so } Q(k) \propto k P(k)$$

Let's check if $\sum_k Q(k) = 1$

$$\sum_k \frac{k}{\langle k \rangle} P(k) = \frac{1}{\langle k \rangle} \sum_k k P(k) = \frac{1}{\langle k \rangle} \cdot \langle k \rangle = 1 \qquad \square$$

p. 7.5 1.5P

A graph is percolated if the percolation cluster's size is equal to $N^* \propto N$. Let's explore the graph in a following way

- pick a random edge in the network and consider one of its end vertices
- the probability that this vertex has degree k is

$$Q(k) = \frac{k}{\langle k \rangle} P(k)$$

consider a vertex reached along a random edge, which has degree k

the number of new vertices reachable from it is $k-1$

the expected value of new vertices per step is

$$\sum_k (k-1) Q(k)$$

If on average, each vertex adds more then one new vertex

$$\sum_k (k-1) Q(k) > 1$$

then starting from a single edge, the exploration produces more and more new vertices at each step

as the network size $N \to \infty$, this leads to a cluster size whose size is proportional to N $\quad N^* \propto N$

so it's a percolated network

percolation threshold

$$\sum_k (k-2) Q(k) \geq 1 \Rightarrow \sum_k k \cdot Q(k) - \sum_k Q(k) \geq 1$$

$$\sum_k k \cdot Q(k) \geq 2$$

substitute

$$Q(k) = \frac{kP(k)}{\langle k \rangle} \Rightarrow \sum_k k \cdot \frac{kP(k)}{\langle k \rangle} = \frac{1}{\langle k \rangle} \cdot \sum_k k^2 P(k) =$$

$$= \frac{\langle k^2 \rangle}{\langle k \rangle} \geq 2$$

$\square$

P 7.12 1.5 P

$$h_1(s) = \sum_{k=0}^{s-1} F(k) h_k(s-1) \qquad h_0(s) = \sum_{k=0}^{\infty} P(k) h_k(s)$$

$$H_k(x) = \sum_{s=0}^{\infty} h_k(s) x^s$$

we want to prove:

1. $H_1(x) = x \sum_{k=0}^{\infty} F(k) H_k(x)$ $\qquad$ 3. $H_1(x) = x G_1(H_1(x))$
2. $H_k(x) = [H_1(x)]^k$ $\qquad$ 4. $H_0(x) = x G_0(H_1(x))$

where $G_0(x) = \sum_{k=0}^{\infty} P(k) x^k$ and $G_1(x) = \sum_{k=0}^{\infty} F(k) x^k$

ad 1

$$H_1(x) = \sum_{s=0}^{\infty} h_1(s) x^s = \sum_{s=0}^{\infty} \left(\sum_{k=0}^{s-1} F(k) h_k(s-1) \right) x^s =$$

$$= \sum_{s=0}^{\infty} \sum_{k=0}^{s-1} F(k) h_k(s-1) x^s = x \sum_{s=1}^{\infty} \sum_{k=0}^{s-1} F(k) h_k(s-1) x^{s-1} =$$

$s = m+1$

$$= x \sum_{m=0}^{\infty} \sum_{k=0}^{m} F(k) h_k(m) x^m = x \sum_{k=0}^{\infty} F(k) \sum_{m=k}^{\infty} h_k(m) x^m$$

$H_k(x) = \sum_{m=k} h_k(m) x^m$ so $H_1(x) = x \cdot \sum_{k=0}^{\infty} F(k) \cdot H_k(x)$

explanation: since we go through k edges, we must reach at least one vertex per edge. Therefore $m \geq k$ otherwise the probability is 0

ad 2

starting from k edges is independent of each other so

$$H_k(x) = H_1(x) \cdot \ldots \cdot H_1(x) \leftarrow k \text{ times}$$

$$H_k(x) = [H_1(x)]^k$$

ad 3

from 1 and 2

$$H_1(x) = x \sum_{k=0}^{\infty} F(k) H_k(x) = x \sum_{k=0}^{\infty} F(k) [H_1(x)]^k = x G_1(H_1(x))$$

ad 4

$H_0(x) = \sum_{s=0}^{\infty} h_0(s) x^s = \sum_{s=0}^{\infty} \sum_{k=0}^{\infty} P(k) h_k(s) x^s = \sum_{k=0}^{\infty} P(k) \sum_{s=0}^{\infty} h_k(s) x^s =$

$= \sum_{k=0}^{\infty} P(k) \cdot H_k(x) = \sum_{k=0}^{\infty} P(k) \cdot [H_1(x)]^k = G_0(H_1(x))$

P 7.6

for ER graphs, the degree distribution can be described as Poisson

$P(k) = \frac{e^{-\langle k\rangle} \langle k\rangle^k}{k!}$ The moments are from previous projects $\langle k\rangle = \langle k\rangle$, $\langle k^2\rangle = \langle k\rangle^2 + \langle k\rangle$

$\frac{\langle k^2\rangle}{\langle k\rangle} = \frac{\langle k\rangle^2 + \langle k\rangle}{\langle k\rangle} = \langle k\rangle + 1$

We are given $\frac{\langle k^2\rangle}{\langle k\rangle} = 2$ so $\langle k\rangle + 1 = 2 \Rightarrow \langle k\rangle = 1$

the mean degree $\langle k\rangle \approx pN$ so $p \approx \frac{1}{N}$ also from previous projects

p 6.3

N nodes (fixed)
edges are added one by one using preferential attachment
node i has degree $k_i(t)$
total degree in the network at time t

$\sum_{j=1}^{N} k_j(t) = 2t$ (handshaking lemma)

Goal: Find probability that a node has degree k at time t

node gains an edge with probability $\frac{k_i}{2t}$
master equation
$p_i(k, t+1) = \frac{k-1}{2t} p_i(k-1, t) + (1 - \frac{k}{2t}) p_i(k, t)$

node had k-1 and gains — node had k and doesn't gain

generating function $G_i(z,t) = \sum_{k=0}^{\infty} p_i(k,t) z^k$

$\langle k\rangle = \frac{\partial G}{\partial z}\Big|_{z=1}$ $\langle k^2\rangle = \frac{\partial^2 G}{\partial z^2}\Big|_{z=1} + \langle k\rangle$

for large t

$p_i(k, t+1) - p_i(k,t) = \frac{k-1}{2t} p_i(k-1,t) - \frac{k}{2t} p_i(k,t) \approx \frac{\partial p_i(k,t)}{\partial t}$

$\frac{\partial p_i(k,t)}{\partial t} = \frac{k-1}{2t} p_i(k-1,t) - \frac{k}{2t} p_i(k,t)$

$\sum_{k=0}^{\infty} \frac{k-1}{2t} p_i(k-1,t) z^k = \sum_{m=-1}^{\infty} \frac{m}{2t} p_i(m,t) z^{m+1} = \sum_{m=0}^{\infty} \frac{m}{2t} p_i(m,t) z^{m+1} =$

$= \frac{z}{2t} \sum_{m=0}^{\infty} m \cdot p_i(m,t) z^m = \frac{z}{2t} \cdot \frac{\partial G_i}{\partial z}$

$-\sum_{k=0}^{\infty} \frac{k}{2t} p_i(k,t) z^k = -\frac{1}{2t} \sum_{k=0}^{\infty} k \cdot p_i(k,t) z^k = -\frac{1}{2t} \cdot \frac{\partial G_i}{\partial z}$

$\frac{\partial G_i}{\partial t} = \frac{z}{2t} \cdot \frac{\partial G_i}{\partial z} - \frac{1}{2t} \frac{\partial G_i}{\partial z} = \frac{z-1}{2t} \frac{\partial G_i}{\partial z}$

$\frac{dG_i}{dt} = \frac{\partial G_i}{\partial t} + \frac{\partial G_i}{\partial z} \frac{dz}{dt}$

we want to find paths $z(t)$ so that G_i doesn't change along the path

$\frac{dG_i}{dt} = 0$

$$0 = \frac{\partial G_i}{\partial t} + \frac{\partial G_i}{\partial z}\frac{dz}{dt} = \frac{\partial G_i}{\partial z} \cdot \left(\frac{z-1}{2t}\right) + \frac{\partial G_i}{\partial z}\frac{dz}{dt}$$

$$\frac{dz}{dt} = -\frac{z-1}{2t}$$

$$\frac{dz}{dt} = \frac{1-z}{2t} \qquad \frac{dz}{1-z} = \frac{dt}{2t} \quad \Big| \int$$

$$\ln|1-z| = \tfrac{1}{2}\ln t + C$$

$$|1-z| = e^{C}\, t^{1/2}$$

16.12.2025 project

www.math.fsu.edu/~bertram/lectures/Diffusion.pdf
(Diffusion on a Graph)

P 9.6

Let c be the diffusion rate across the edge, then the amount of substance that moves from node j to node i over a time period dt is $c(u_j - u_i)\,dt$ and from node i to node j is $c(u_i - u_j)\,dt$. So

$$\frac{du_i}{dt} = c(u_j - u_i) \qquad \frac{du_j}{dt} = c(u_i - u_j)$$

Diffusion to and from node i must take into consideration all nodes in the graph. The connectivity of the graph is encoded in the adjacency matrix.

$$\frac{du_i}{dt} = cA_{i1}(u_1 - u_i) + cA_{i2}(u_2 - u_i) + \dots + cA_{in}(u_n - u_i) \quad \text{or}$$

$$\frac{du_i}{dt} = c\sum_{j=1}^{n} A_{ij}(u_j - u_i)$$

$$\frac{du_i}{dt} = c\sum_{j=1}^{n} A_{ij}u_j - cu_i \underbrace{\sum_{j=1}^{n} A_{ij}}_{\text{degree of node } i \ d_i} = c\sum_{j=1}^{n} A_{ij}u_j - cu_i d_i$$

$$\delta_{ij} = \begin{cases} 0 & \text{if } i \neq j \\ 1 & \text{if } i = j \end{cases}$$

$$cu_i d_i = c\sum_{j=1}^{n} \delta_{ij} u_j d_j$$

$$\frac{du_i}{dt} = c\sum_{j=1}^{n} A_{ij}u_j - c\sum_{j=1}^{n} \delta_{ij} u_j d_j \qquad u = \begin{bmatrix} u_1 \\ \vdots \\ u_n \end{bmatrix}$$

$$c\sum_{j=1}^{n} A_{ij}u_j = c[Au]_i$$

D - degree matrix

A - adjacency matrix

$D = \mathrm{diag}(d_1, \dots, d_n)$

$$c\sum_{j=1}^{n} \delta_{ij} u_j d_j = c[Du]_i$$

$$\frac{du}{dt} = cAu - cDu = c(A - D)u$$

Graph Laplacian

$L \equiv D - A$

$$\frac{du}{dt} + c(D - A)u = 0$$

The equation for diffusion on a graph is then

$$\frac{du}{dt} + cLu = 0$$

$$L_{ij} = \begin{cases} d_i & \text{if } i = j \\ -1 & \text{if } i \neq j \text{ and there is an edge} \\ 0 & \text{if } i \neq j \text{ and there is no edge} \end{cases}$$

L is symmetric so it has real eigenvalues and orthogonal eigenvectors v_i

solution as linear combination of eigenvectors

$$u = \sum_{i=1}^{n} a_i(t)\, v_i$$

$a_i(t)$ time dependent coefficient along this eigenvector

$$\sum_{i=1}^{n} \frac{da_i}{dt} v_i + \sum_{i=1}^{n} c a_i L v = 0 \Rightarrow \sum_{i=1}^{n} \left(\frac{da_i}{dt} + c a_i \lambda_i \right) v_i = 0$$

λ_i - eigenvalue of L

The eigenvectors v_i form an orthogonal set, meaning no eigenvector is a linear combination of the others

For a sum of orthogonal vectors equal to zero, we must have each coefficient equal to zero individually

$\frac{da_i}{dt} + c\lambda_i a_i = 0 \quad i = 1, \dots, n$ first order linear differential equation

so the solution is $a_i(t) = a_i(0)\, e^{-c\lambda_i t}$ $a_i(0)$ initial value of the coefficient

Since each coefficient has such a solution, then by the superposition principle, a linear combination of these is also a solution.

$u(t) = \sum_{i=1}^{n} a_i(0)\, e^{-c\lambda_i t}\, v_i$

$u(0)$ initial values on each node

how to find $a_i(0)$?

$u(0) = \sum_{i=1}^{n} a_i(0)\, v_i$

$u(0) \cdot v_j = a_j(0)\, |v_j|^2 \Rightarrow a_j(0) = \frac{u(0) \cdot v_j}{|v_j|^2}$

L is singular because:

Look at any row i. The diagonal element is the degree of the node d_i. All the other elements are either 0 or for each edge -1.

There are exactly d_i of these, so if you sum across any row of L you get $d_i - d_i = 0$. This is true for any of the rows.

so the sum of all columns of the matrix is 0. Therefore, L is singular

That is, it has at least one zero eigenvalue. $\lambda_1 = 0$

Does L have any negative eigenvalues?

Suppose that $\lambda_2 < 0$. Then the term in the spectral solution

$a_2(0)\, e^{-c\lambda_2 t}\, v_2 \to \infty$ as $t \to \infty$ but concentration in one node can't go to ∞

So L has non-negative eigenvalues

The number of 0 eigenvalues of the graph Laplacian is equal to the number of components of the graph.

In order for diffusion to reach a single global state the graph needs to be connected.

λ_2 algebraic connectivity

so $\lambda_1 = 0$ $\quad 0 = \lambda_1 < \lambda_2 \leq \dots \leq \lambda_N$

Remember

$u(t) = \sum_{i=1}^{n} a_i(0)\, e^{-c\lambda_i t}\, v_i$

Diffusion speed is dominated by the slowest decaying part in the sum which is $a_2(0)\, e^{-c\lambda_2 t}\, v_2$ because it's constant for 0

larger $\lambda_2 \Rightarrow$ faster convergence to steady state

so the graph with fastest diffusion has maximum λ_2 given n nodes. let's find one

Wikipedia Rayleigh quotient

$R(x) = \frac{x^T L x}{x^T x}$ for non-zero vector x

For symmetric L, the Rayleigh quotient is minimized by the eigenvector corresponding to the smallest eigen vectors

The k-th eigenvalue can be obtained by minimizing $R(x)$ over all vectors orthogonal to the first $k-1$ eigenvectors

eigenvector corresponding to λ_1 it's a vector of 1s. why?

Because $L\mathbb{1}$ is the sum of the columns of L, which we know equals 0. So

$Lv_1 = 0$

To find λ_2 we look at the minimum of the Rayleigh quotient over all vectors orthogonal to $\mathbb{1}$

$$\lambda_2 = \min_{x \perp \mathbb{1}} \frac{x^T L x}{x^T x}$$ so we ignore $\lambda_1 = 0$

$$x^T L x = x^T D x - x^T A x = \sum_{i=1}^{n} d_i x_i^2 - \sum_i \sum_j 2 x_i x_j$$

$$(x_i - x_j)^2 = x_i^2 + x_j^2 - 2x_i x_j$$

$$\sum_{i,j} (x_i - x_j)^2 = \sum_{i,j} (x_i^2 + x_j^2 - 2x_i x_j)$$

Each x_i^2 appears once for each neighbor so total contribution $= d_i x_i^2$

Therefore $x^T L x = \sum_{i,j} (x_i - x_j)^2$

Focusing on a node with minimum degree — $N(i)$ - neighbors of i

suppose node i has degree d_i $\quad \sum_{j \in N(i)} (x_i - x_j)^2$

$$\sum_{j \in N(i)} (x_i - x_j)^2 \geq \sum_{j \in N(i)} (x_i^2 + x_j^2) = d_i x_i^2 + \sum_{j \in N(i)} x_j^2$$

So $\sum_{j \in N(i)} (x_i - x_j)^2 \geq d_{min} x_i^2$

$$\sum_{i,j} (x_i - x_j)^2 \geq d_{min} \sum_i x_i^2 = d_{min} x^T x$$

$$\lambda_2 = \min_{x \perp \mathbb{1}} \frac{x^T L x}{x^T x} \Rightarrow \lambda_2 \leq d_{min}$$

So the node with the fewest neighbors is the bottleneck limiting the algebraic connectivity.

A graph with a fixed number of edges and vertices which has the highest d_{min} is:

$\boxed{\text{complete graph}}$

which makes sense intuitively.

P 9.1 www.lirmm.fr/~sau/JCALM/Josep.pdf Random Walks on Graphs

Let A be the adjacency matrix of a connected undirected graph with n nodes. Define a random walk with transition probabilities

$p_{ji} = \frac{A_{ij}}{k_i}$ $k_i = \sum_j A_{ij}$ (degree of node i)

stationary distribution

for undirected graphs

$\pi_j = \sum_i \pi_i p_{ji}$ $\sum_i \pi_i = 1$

$A_{ij} = A_{ji}$

$\pi_i \frac{A_{ij}}{k_i} = \pi_j \frac{A_{ji}}{k_j} = \pi_j \frac{A_{ij}}{k_j}$ $A_{ij} = A_{ji}$

flow from i to j = flow from j to i

$\pi_i p_{ji} = \pi_j p_{ij}$

$\Rightarrow \frac{\pi_i}{k_i} = \frac{\pi_j}{k_j}$ so $\pi_i \propto k_i$ $\forall_{ij}$ $A_{ij} = 1$

$k_i \neq 0$ because graph is connected

but we need to normalize in order to get distribution

since $\sum_i \pi_i = 1$

$c = \frac{\pi_i}{k_i}$

$\sum_{i=1}^n \pi_i = \sum_{i=1}^n c \cdot k_i = c \sum_{i=1}^n k_i = 1 \Rightarrow c = \frac{1}{\sum_{i=1}^n k_i}$

$\pi_i = \frac{k_i}{\sum_{j=1}^n k_j}$

from lecture

Let us consider non-periodic and irreducible Markov chain with finite number of states. Then this chain has exactly one stationary distribution π. Moreover, distribution π is ergodic distribution of this process.

The chain is irreducible because graph is connected so there exists a path between any two nodes.

Therefore, starting from any node i, it is possible to eventually reach any other node j by following edges.

the graph will be non-periodic if it has a self-loop or isn't bipartite.